

PIERRE-LOUIS SIXDENIER

PhD Candidate in Computer Science & Artificial Intelligence

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EDUCATION

Doctor of Philosophy (Ph.D.) in Computer Science

Vision & Learning Laboratory • Doctoral School

2023 – Present (Exp. 2026)

Paris, France

- **Thesis:** Scalable Self-Supervised Representations for Multimodal Perception in Low-Resource Regimes
- **Advisors:** Prof. Jane Doe, Dr. Alex Martin • **Honors:** Full National Doctoral Research Fellowship

M.Sc. in Computer Science & Artificial Intelligence

Institute of Technology • Faculty of Sciences

2021 – 2023

France

- **Graduation:** Summa Cum Laude (Rank 1st / 85, GPA: 4.0/4.0) • **Thesis:** Geometric Deep Learning for 3D Segmentation

B.Sc. in Mathematics and Computer Science

University Faculty of Exact Sciences

2018 – 2021

France

- **Graduation:** First-Class Honors (Mention Très Bien) • Dean's Honor List across all semesters

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (PhD Candidate)

Vision & Learning Laboratory

Oct 2023 – Present

Paris, France

- Investigated self-supervised pre-training objectives for visual representation learning and multimodal reasoning.
- Scaled distributed training workflows across 64+ NVIDIA H100 GPUs using PyTorch FSDP and DeepSpeed.
- Mentored 4 Master's and undergraduate students on machine learning research projects.

Research Scientist Intern

AI Research Labs (Multimodal Foundations)

May 2024 – Sep 2024

London, UK / Hybrid

- Developed parameter-efficient fine-tuning and contrastive distillation algorithms for vision transformers.
- Reduced inference latency by 38% while retaining 99.2% zero-shot accuracy across 12 vision benchmarks.
- Filed 1 patent on token pruning mechanisms for streaming multimodal architectures.

Graduate Teaching Fellow

Department of Computer Science

Sep 2022 – Jun 2023

University Campus

- Headed lab recitations for CS-401 Deep Learning (60 students) and CS-202 Algorithms & Data Structures (120 students).

SELECTED SCIENTIFIC PUBLICATIONS

[1] Contrastive Representation Learning with Adaptive Geometry for Low-Resource Multimodal Tasks

Pierre-Louis Sixdenier, Jane Doe, Alex Martin.

Advances in Neural Information Processing Systems (*NeurIPS 2025*) – Spotlight

[2] Token-Sparse Vision Transformers: Adaptive Computation for Real-Time Dense Prediction

Lucas Bernard, Pierre-Louis Sixdenier, Elena Rossi, Jane Doe.

IEEE/CVF Conference on Computer Vision and Pattern Recognition (*CVPR 2025*) – Oral Presentation

[3] Provable Convergence Bounds for Gradient-Based Self-Distillation in Deep Ensembles

Pierre-Louis Sixdenier, Alex Martin.

International Conference on Learning Representations (*ICLR 2024*)

SIDE PROJECTS & SOFTWARE TOOLING

- **FastVisionBench:** High-throughput benchmarking harness for few-shot representation quality across 18 datasets in <15 min. (Python, PyTorch, Distributed)
- **LatentManifold.js:** WebAssembly & WebGL interactive 3D browser visualizer for t-SNE, UMAP, and PCA in real time. (TypeScript, Three.js, Wasm)
- **AutoBibSync:** Automated CI tool synchronizing DBLP, arXiv, and Google Scholar into validated BibTeX collections. (Python CLI, GitHub Actions)

TECHNICAL SKILLS & ACADEMIC SERVICE

Technical Stack: Python (Expert), C/C++, CUDA, PyTorch, JAX/Flax, DeepSpeed, FSDP, Linux, SLURM, Docker, Git, LaTeX.

Academic Service: Reviewer for *NeurIPS* (2024, 2025), *CVPR* (2025), *ICLR* (2025); ECCV Workshop Co-Organizer.